that if safety alignment enables the model to consistently re-select the correct reasoning direction at each step (by re-evaluating the query and previously generated tokens before producing the next one), the model can continue to generate outputs that are both safe and helpful, even confronted with adversarial attempts. This perspective has been validated explicitly by a recent study, which demonstrates its practical effectiveness in mitigating jailbreak attacks (Li & Kim, 2025). This also suggests that improving local consistency (safety) in reasoning can be a key pathway toward building more robust and attack-resilient language models.

Challenges in Proving. While **SSAH** provides a more specific focus than SAH, empirically proving it still presents significant challenges. A key issue is the infeasibility of sampling sufficient outputs to fully capture the model's distribution of responses across both *safety-aligned* and *non-safety-aligned* models. This challenge makes it hard to draw comprehensive and profound conclusions or interpret certain model behaviors solely from benchmark outputs.

However, we approach the problem from an alternative perspective: if SSAH holds, we should observe distinct and consistent differences in the reasoning direction at each step of generation between safety-aligned and non-safety-aligned models. In a safety-aligned model, the reasoning direction should consistently guide the model in rejecting harmful queries at every token generation step. In contrast, a non-safety-aligned model might exhibit reasoning patterns that incline toward fulfilling malicious requests. **Rather than relying solely on surface-level benchmark evaluations, we can probe the model's reasoning direction to gain deeper insights into its internal decision-making process at each step, regardless of the specific outputs produced**.

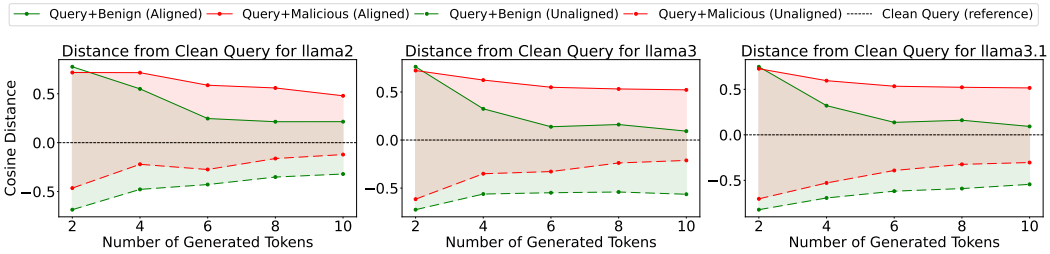

Figure 2: Probing reasoning direction on the AdvBench dataset with **Llama2-7B**, **Llama3-8B**, and **Llama3.1-8B** using cosine distance. Models were finetuned to ensure that aligned versions possess both general instruction-following abilities and safety guardrails, while unaligned models only have instruction-following capabilities. More results are in Appendix B.2.

Probing Experiment. Although we cannot directly observe the model's reasoning direction, as a turnaround, we can infer it by measuring the *distance* between hidden states in feature space when the model is fed with queries that follow different reasoning paths. By comparing the distances in aligned and unaligned models, we gain insights into how safety alignment affects the model's reasoning direction in each generation step. Specifically, we observe that the model's behavior can be influenced by the initial tokens in its response. For instance, appending certain tokens to a malicious query can lead an *aligned model* to produce unsafe responses or prompt an *unaligned model* to generate safe ones. This indicates that *initial response tokens* can alter the model's reasoning direction. Based on this observation, we construct three types of queries to probe the model's reasoning trajectory (more details in Appendix B.2): **(1) Clean Query**: The original malicious query (e.g., "How to make a bomb?"); **(2) Query + benign tokens**: The malicious query followed by benign

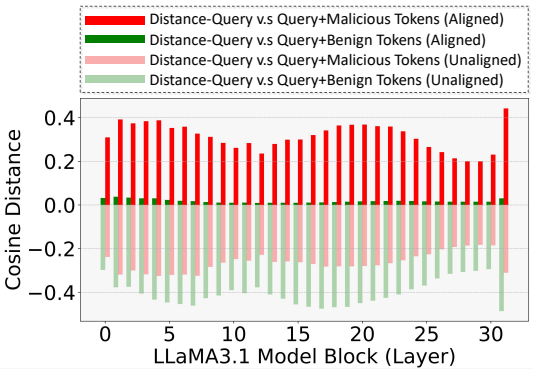

Figure 3: Cosine distance of hidden states between various crafted and clean queries across all blocks of LLMs (Aligned/Unaligned definitions are the same as Fig. 2).

prompt token (e.g., "How to make a bomb? Sorry, I can't..."); **(3) Query + malicious tokens**: The malicious query followed by malicious prompt tokens (e.g., "How to make a bomb? Here's how...").

Expected Outcomes and Probe Results. For an *aligned model*, we expect the *hidden state distances* between the **query** and **query + benign token** to be shorter than those between the **query** and **query + malicious token** at each generating step. In contrast, for an *unaligned model*, we anticipate the opposite: the distances between the **query** and **query + malicious token** should be shorter than those between the **query** and **query + benign token**. If these patterns are observed, it indicates that safety alignment has successfully established the model's ability to choose the correct reasoning direction. This also suggests that safety alignment reshapes the model's internal decision-making process, ensuring safer behavior from the very beginning of the response. As shown in Fig. 2, the probe results provide evidence that safety alignment teaches the model's correct reasoning direction as hypothesized. **However, it is important to note that this evidence is necessary but not enough, as safety alignment may introduce more nuanced changes that are not fully captured by SSAH**.

Results Analysis and Discussion. We also present the aforementioned distances and their *differences* across Transformer blocks in Fig. 3 and Fig. 4, respectively. Our findings demonstrate that the previous conclusions are consistently upheld across all Transformer blocks. Specifically, the aligned model shows larger distance differences compared to the unaligned model, suggesting that the unaligned model lacks a strong preference for safe to unsafe reasoning. In contrast, the aligned model exhibits a clear preference for safe reasoning, as reflected by the more pronounced distance differences. Moreover, we observe that in the unaligned model, the distance difference gradually increases across the earlier Transformer blocks 0 - 7. However, in the aligned model, the distance difference remains consistently large throughout all blocks. This indicates that the preference for safe reasoning in the aligned model is embedded not only in the later layers (which typically capture higher-level features), but also in the earlier layers. Consequently, safety alignment influences the model's internal decision-making from the initial stages of processing.

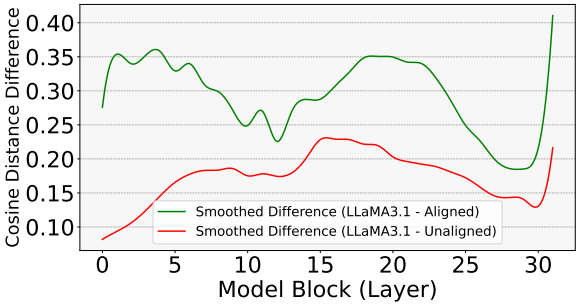

Figure 4: Absolute ***differences*** of cosine distance of Fig. 3 across all blocks of LLMs: *Abs(Distance(Query + Benign tokens, Query) - Distance(Query + Malicious tokens, Query))*.

4 Less is More for Safety Alignment

While the full validation of **SSAH** remains challenging due to the high complexity of the output sampling space, we are able to empirically validate a key implication. Specifically, if **SSAH** holds, then **safety alignment can be achieved with only a small subset of critical computing units**, as the task can be interpreted as an implicit and simple safety-related binary classification problem.

4.1 Identifying Safety-Critical Computing Units

To verify this implication, we designed experiments to determine the minimally essential subset of computing units in LLMs that is critical in establishing a safety guardrail. Following this, we hypothesize that specific attributes of LLMs can be explicitly linked to certain computing units within the model. Our experiments are designed as follows:

Definition of Attribute Groups. This paper considers two attributes of LLMs, *utility* and *safety*. We first exclusively link safety or utility attributes to specific computing units. We also speculate some units may contribute to both attributes simultaneously. Moreover, we hypothesize that certain computing units are redundant and do not associate with any attribute. Therefore, we categorize the computing units of LLMs into four groups: ***Safety Critical Units (SCU)*** and ***Utility Critical Units (UCU)***, corresponding exclusively to either safety and utility, respectively; ***Complex Units (CU)***, contributing to both safety and utility; and ***Redundant Units (RU)***, not associated with any attribute.

Table 1: Pruning results of **Llama2-7B-Chat** and **Llama3-8B-Instruct** across safety and utility benchmarks. Breakdown of model performance after pruning various categories of computing units, including SCU, UCU, and RU, demonstrating their respective contributions to safety and utility attributes. The proportion of each attribute group in the model is determined based on the degradation in utility and safety. Additional evaluation details are provided in Appendix C.1 and C.2.

Type	wiki2	Utility (ACC%)							Safety (ASR %)		
		wino	openb	arc_c	boolq	hellas	rte	avg	w/ sys	w/o sys	avg
Meta-Llama-2-7B-Chat											
Dense	6.49	65.5	32.5	43.5	79.5	57.0	71.5	58.3 (-0)	3.0	18.0	10.0 (+0)
SCU (**1.3%**)	6.76	64.0	34.0	42.5	78.5	52.0	70.5	56.9 (-1.3)	19.0	84.0	**66.0 (+56.0)**
UCU (13.3%)	180.2	56.5	22.5	25.0	59.5	36.5	56.0	**42.7 (-15.6)**	23.0	48.0	28.3 (+18.3)
RU (14.8%)	8.32	63.5	34.5	39.0	75.5	55.5	64.5	55.5 (-2.8)	6.0	19.0	14.6 (+4.6)
Meta-Llama-3-8B-Instruct											
Dense	7.74	71.5	34.5	51.0	80.0	60.0	70.0	61.2 (-0)	2.0	29.0	15.5 (+0)
SCU (**1.4%**)	9.06	67.0	30.5	45.0	82.0	55.5	65.5	57.6 (-3.6)	80.0	93.0	**86.5 (+71.0)**
UCU (6.8%)	269.2	60.0	23.0	25.0	59.5	47.5	51.5	**44.4 (-16.8)**	16.0	24.0	20 (+4.5)
RU (6.6%)	8.52	74.0	31.0	50.5	80.0	57.5	71.5	60.8 (-0.4)	1.0	24.0	12.5 (-3.0)

Verfication of Attribute Group. To verify our hypothesis that different groups of computing units contribute exclusively, collectively, or neither to safety and utility attributes, we use a model pruning mechanism. The rationale behind pruning is that removing components most closely linked to a specific attribute would significantly impact the model's performance in that area - it is a sort of ablation study. As pruning reduces the model's capacity, the most affected attributes reveal the critical components for that function.

Following Wei et al. (2024), we construct two datasets to separately evaluate the model's performance on utility and safety. The utility dataset measures the model's functional capabilities (e.g., general reasoning, language understanding), while the safety dataset evaluates its ability to reject harmful or unethical queries. This allows us to identify the computing units most closely associated with utility and safety, respectively. Unlike previous approaches that identify safety-critical components at the weight level, we identify them at the ***neuron level***, focusing on individual neurons and channels within the model. Specifically, we use a structured pruning strategy inspired by An et al. (2024), which removes structured components of each **depth-2** module based on the variance of activation values across a target dataset. Given a depth-2 module, $f(X) = B\sigma(AX)$, which can represent either an attention module or a feedforward module, where A and B are weight matrices. Then, we define the importance score for each channel as follows:

$$\mathbf{I}_{:,j} = \frac{1}{N-1} \sum_{n=1}^{N} \left(X^B_{n,j,:} - \overline{X}^B_{:,j,:} \right)^2 \cdot \|\mathbf{W}^B_{:,j}\|_2^2 \tag{1}$$

Here, N refers to the number of calibration samples, $\mathbf{W}^B_{:,j}$ refers to the j-th column of B, and X^B represents the input to B. With this score, we plan to prune the input channels of B and the output neurons of A, as channels or neurons with low activation variance across the target dataset are considered less important - details in Appendix C.3.

In this way, we calculate the *importance score* for each individual neuron or channel, denoted as $\mathbf{I_U}$ for the utility attribute and $\mathbf{I_S}$ for the safety attribute. Initially, we prune the computing units with the smallest $\mathbf{I_U} + \mathbf{I_S}$ values to identify redundant units. Subsequently, we prune units with the largest and smallest $\mathbf{I_S} - \mathbf{I_U}$ values to identify utility and safety critical units, respectively. The remaining computing units are categorized as complex units. We experiment with various pruning ratios (starts with higher ratios and systematically reduces them) and evaluate the resulting safety and utility performance, selecting the optimal pruning ratio with minimal performance degradation in the corresponding attribute. This systematic pruning process enables us to validate our hypothesis and derive the roles of different units accurately.

Ablation Study Results. The experiment results are described in Tab. 1. We discovered that for a safety-aligned model, the computing units that are exclusively responsible for the safety attribute account for only about **1.3 - 1.4%** of the total units. Although the complex units make up a larger portion of the model, their primary role is to support the general knowledge required for both safety